

Clinical Decision Support System

MRI Brain Tumor Analysis Report

Patient ID	ANON_NOTUMOR
Date/Time	2026-05-18T09:59:41.249942
Run ID	20260518_095936
Tumor Type	NOTUMOR
Confidence	100.0%
Radiologist	Approved

Clinical Summary

No intracranial neoplasm identified. Findings are within normal limits. Clinical correlation with presenting symptoms is recommended.

Recommended Next Steps

1. Reassure patient — no intracranial neoplasm identified
2. Clinical correlation with presenting symptoms
3. Consider alternative diagnoses (migraine, demyelination, vascular)
4. Repeat imaging only if symptoms progress or new deficits emerge

Clinical Guidelines

NO TUMOR

Definition: No intracranial neoplasm identified. Brain parenchyma and extra-axial spaces are within normal limits.

Normal MRI Findings:

- Normal gray-white matter differentiation
- Symmetrical cortical sulci and gyri without abnormal signal
- Normal ventricular size (no hydrocephalus)
- No focal mass, abnormal enhancement, or mass effect
- Normal pituitary gland, no sellar mass
- Intact blood-brain barrier (no abnormal contrast enhancement)

Clinical Significance:

- Reassuring result — excludes structural neoplastic pathology
- Presenting symptoms are attributable to non-neoplastic causes
- Requires clinical correlation for further workup

Recommendations:

1. Reassure patient: no evidence of intracranial tumor
 2. Clinical correlation with presenting symptoms is essential
 3. Consider alternative diagnoses:
 - Migraine or tension headache
 - Demyelinating disease (MS) — check for FLAIR white matter lesions
 - Vascular causes (review MRA/MRV if available)
 - Psychiatric or functional neurological disorder
 4. Repeat imaging only if symptoms progress, new neurological deficits emerge, or clinical suspicion remains high
 5. Consider other modalities if indicated: CT for calcification, FDG-PET for metabolic activity, EEG for epilepsy workup
 6. Neurology referral if symptoms persist without clear alternative explanation
- GLIOMA

Definition: Glioma is a type of tumor that starts in the glial cells of the brain or spine. Glial cells support and protect neurons.

Characteristics:

- Fast-growing, often malignant tumors
- Can originate anywhere in the brain or spinal cord
- Most common primary brain tumors in adults (approximately 30% of all brain tumors)
- Graded I-IV based on aggressiveness (WHO classification); Grade IV (glioblastoma) is the most aggressive
- Often infiltrative, making complete surgical resection difficult

Typical Imaging Findings (MRI):

- Irregular, poorly defined borders
- Heterogeneous signal intensity on T1/T2 sequences
- Ring enhancement...

This report is generated by an AI-assisted clinical decision-support system. It is NOT a substitute for professional medical diagnosis. All findings must be confirmed by a qualified radiologist or physician.